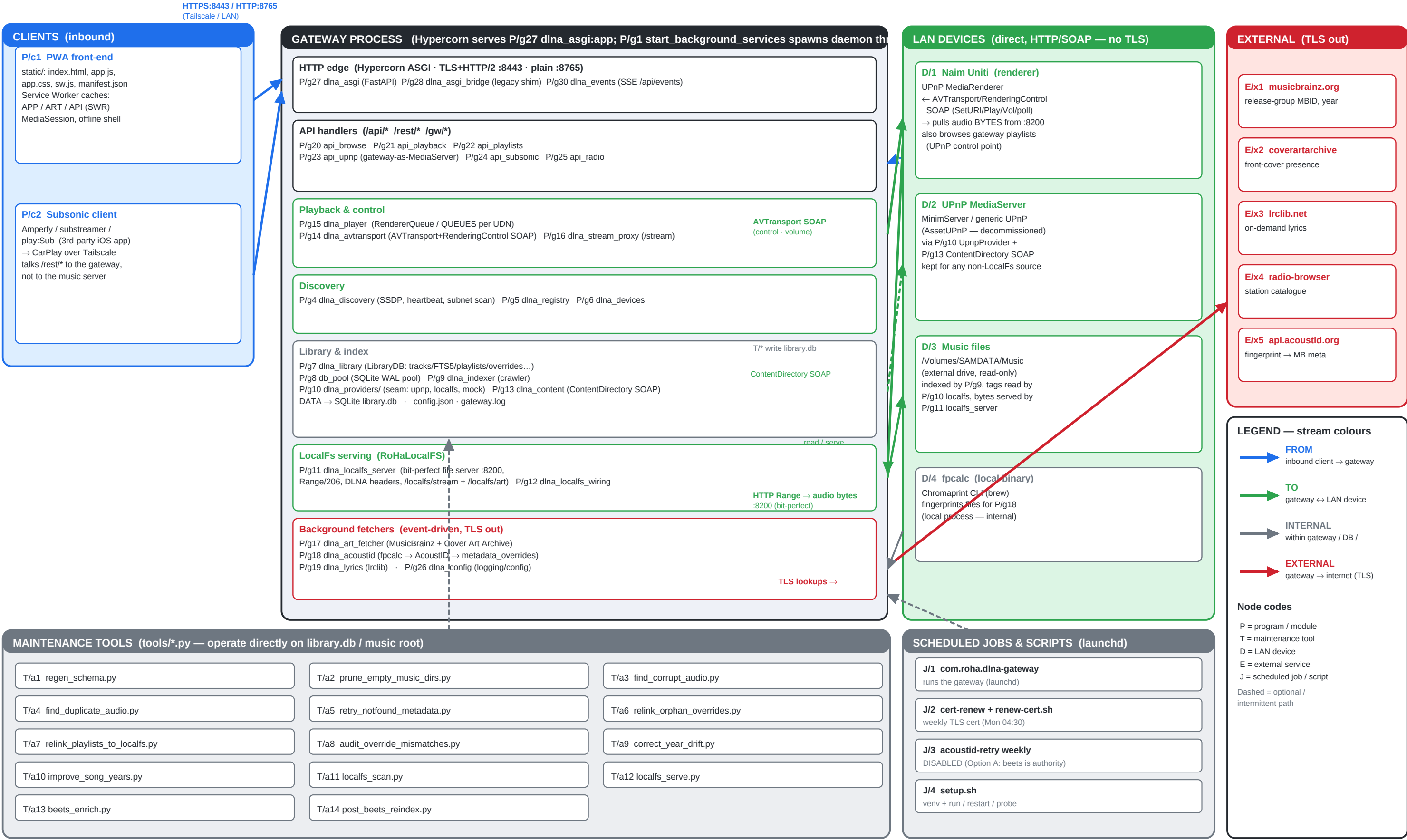


DLNA Gateway — Architecture (2.0 · ASGI/Hypercorn)

What runs where · stream colours show direction/scope · node codes (P/T/D/E/J) index into the lists overleaf



Node lists — every box in the drawing

Codes match the diagram. **P** = program / module, **T** = tool, **D** = device, **E** = external service, **J** = scheduled job.

Application programs & modules (P/\*)

| Code  | File / location                                          | Responsibility                                                                                                                                                                                                                                                                                                                                               |
|-------|----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P/c1  | static/index.html, app.js, app.css, sw.js, manifest.json | PWA front-end. Browse/playback UI, MediaSession lock-screen, Service-Worker caches (APP shell SWR, ART cache-first, API stale-while-revalidate), offline shell.                                                                                                                                                                                              |
| P/c2  | (3rd-party iOS app — Amperfy / substreamer / play:Sub)   | Subsonic client for CarPlay over Tailscale. Talks /rest/* to the gateway; never to the music server directly.                                                                                                                                                                                                                                                |
| P/g1  | dlna_gateway.py                                          | Module-wiring + start_background_services(): spawns the daemon threads (SSDP listener, pre-prober, subnet scanner, heartbeat, gateway announcer). Called from the dlna_asgi lifespan so `hypercorn dlna_asgi:app` boots the whole gateway. Its own stdlib HTTP edge + TLS were removed in 2.0 (Hypercorn owns the edge).                                     |
| P/g2  | dlna_server.py                                           | 2.0: reduced to the DEVICE-TIER server — plain-HTTP /gw/* for the Naim (DeviceHandler/start_device_server) on :8770 (GATEWAY_PORT), started by the ASGI lifespan alongside Hypercorn. The old main HTTP edge role moved to P/g27. (Cleanup C will fold /gw into the ASGI app and retire this.)                                                               |
| P/g3  | dlna_routes.py                                           | GET_ROUTES / POST_ROUTES path→handler maps; the ASGI bridge (P/g28) mounts the not-yet-native ones.                                                                                                                                                                                                                                                          |
| P/g4  | dlna_discovery.py                                        | SSDP multicast listener, probe, subnet scanner, renderer/server heartbeat. Holds SERVERS / RENDERERS singletons.                                                                                                                                                                                                                                             |
| P/g5  | dlna_registry.py                                         | Data classes + thread-safe ServerRegistry / RendererRegistry.                                                                                                                                                                                                                                                                                                |
| P/g6  | dlna_devices.py                                          | DeviceRoleCache — in-memory mirror of device_roles for zero-latency classification.                                                                                                                                                                                                                                                                          |
| P/g7  | dlna_library.py                                          | LibraryDB — SQLite index, FTS5 search, playlists, album_art, play_counts, lyrics, metadata_overrides, favourites, radio. Composition root for DB-owning singletons + fetchers.                                                                                                                                                                               |
| P/g8  | db_pool.py                                               | SQLite connection pool — WAL, thread-local conns, write serialization.                                                                                                                                                                                                                                                                                       |
| P/g9  | dlna_indexer.py                                          | Background crawler that walks a provider and populates LibraryDB (with FTS self-heal).                                                                                                                                                                                                                                                                       |
| P/g10 | dlna_providers/ (__init__, upnp.py, localfs.py, mock.py) | LibraryProvider seam + registry. UpnpProvider wraps the UPnP SOAP path (kept for MinimServer); LocalFsProvider is the in-process backend (mutagen, content-hashed ids, watchdog).                                                                                                                                                                            |
| P/g11 | dlna_localfs_server.py                                   | RoHaLocalFS bit-perfect file server (:8200). Range-aware (206/416), DLNA-headered, /localfs/stream/ + /localfs/art/. Path-traversal defended.                                                                                                                                                                                                                |
| P/g12 | dlna_localfs_wiring.py                                   | Boot-time wiring of the LocalFs provider (maybe_start_localfs); gated on LOCALFS_MUSIC_ROOT / config.                                                                                                                                                                                                                                                        |
| P/g13 | dlna_content.py                                          | UPnP ContentDirectory SOAP client (cd_browse/cd_search). Reached only via P/g10 upnp.py.                                                                                                                                                                                                                                                                     |
| P/g14 | dlna_avtransport.py                                      | UPnP AVTransport + RenderingControl SOAP (SetURI/Play/Stop/Pause, state/position probe, SetVolume).                                                                                                                                                                                                                                                          |
| P/g15 | dlna_player.py                                           | RendererQueue (sequential per-renderer playback, gapless via SetNextAVTransportURI, watchdog) + QueueRegistry (QUEUES, one per UDN).                                                                                                                                                                                                                         |
| P/g16 | dlna_stream_proxy.py                                     | Browser-audio HTTP proxy /stream (Range pass-through, 5-min idle timeout) + proxy_radio_stream /radio_stream (ICY de-interleave).                                                                                                                                                                                                                            |
| P/g17 | dlna_art_fetcher.py                                      | AlbumArtFetcher — MusicBrainz release-group + Cover Art Archive lookup; sticky notfound cache; ~1 req/s.                                                                                                                                                                                                                                                     |
| P/g18 | dlna_acoustid.py                                         | AcoustIDFetcher — fpcalc fingerprint → AcoustID → MusicBrainz metadata into metadata_overrides; transient-vs-permanent split; year backfill. DORMANT under Option A: beets (T/a13) is now the metadata authority, so ACOUSTID_API_KEY is left unset and this worker no-ops.                                                                                  |
| P/g19 | dlna_lyrics.py                                           | On-demand Irclib lyrics; cached in the lyrics table; sticky positive + negative.                                                                                                                                                                                                                                                                             |
| P/g20 | api_browse.py                                            | Browse / search / radio-shuffle API.                                                                                                                                                                                                                                                                                                                         |
| P/g21 | api_playback.py                                          | Playback, /stream route, /art proxy, /api/client_log, state, indexer + AcoustID management.                                                                                                                                                                                                                                                                  |
| P/g22 | api_playlists.py                                         | Playlist CRUD endpoints.                                                                                                                                                                                                                                                                                                                                     |
| P/g23 | api_upnp.py                                              | UPnP service descriptors + ContentDirectory SOAP exposing the gateway's own playlists + favourites (so the Naim can browse it as a MediaServer).                                                                                                                                                                                                             |
| P/g24 | api_subsonic.py                                          | Subsonic-compatible /rest/* API (browse, playlists, favourites→starred, stream, cover, scrobble, internet radio) for CarPlay clients.                                                                                                                                                                                                                        |
| P/g25 | api_radio.py                                             | Internet-radio endpoints: radio-browser search, favourites (≤25), ICY now-playing.                                                                                                                                                                                                                                                                           |
| P/g26 | dlna_config.py                                           | Constants (DB_FILE/CFG_FILE/LOG_FILE), logging setup, config load/save, .env load (if dotenv present), raise_fd_limit(8192).                                                                                                                                                                                                                                 |
| P/g27 | dlna_asgi.py                                             | 2.0 MAIN EDGE. FastAPI app served by Hypercorn (TLS+HTTP/2 on :8443 via ALPN, plain :8765); owns the tailscale cert. Native routes for the read API, /art, /stream + /radio_stream relays, static/PWA, and the Subsonic byte methods; legacy handlers run via the bridge. Lifespan boots P/g1 services + the P/g2 device server. docs_url off (no CDN call). |
| P/g28 | dlna_asgi_bridge.py                                      | Shim that runs the legacy (h, params) handlers unchanged inside the ASGI app (a fake `h` captures _json/_html/_xml/send_error; runs in a threadpool). Routes are rewritten native one batch at a time, then dropped from the bridge.                                                                                                                         |
| P/g29 | dlna_art_cache.py                                        | On-disk cover-art byte cache keyed by source URL. art_fetch_cached() (in P/g21) fronts art_fetch so /art and Subsonic getCoverArt serve repeat covers from disk (across clients + restarts) instead of re-fetching coverartarchive / re-decoding embedded art. TTL + size capped; art_cache/ gitignored.                                                     |
| P/g30 | dlna_events.py                                           | EventBus/EVENTS — thread-safe publish to the asyncio loop; native GET /api/events (SSE). Publishers: RendererQueue state, index status transitions, discovery changes. The PWA opens an EventSource as a polling accelerator (fallback intact).                                                                                                              |

Maintenance tools (T/\*) — tools/\*.py

Operate directly on library.db and/or the music root. All default to dry-run / report-only; deletions move to Trash unless --hard-delete. Each has a unit-test sibling tools/test\_\*.py.

| Code  | File                          | Purpose & key options                                                                                                                                                                                                                                                                                                 |
|-------|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T/a1  | regen_schema.py               | Regenerate the committed schema.sql artifact after any schema change. --check = fail if stale (the test_schema_sync gate).                                                                                                                                                                                            |
| T/a2  | prune_empty_music_dirs.py     | Trash directories whose subtree has zero music files. --dry-run -v --limit N --exts a,b --hard-delete -y.                                                                                                                                                                                                             |
| T/a3  | find_corrupt_audio.py         | Flag files with bad/zero magic bytes for their extension. Writes corrupt-audio.txt. --trash --hard-delete --limit --exts --out -y.                                                                                                                                                                                    |
| T/a4  | find_duplicate_audio.py       | Find duplicate recordings on disk (same acoustid metadata); ranks a winner by bit-depth/sample-rate/size. --trash --hard-delete -v -y.                                                                                                                                                                                |
| T/a5  | retry_notfound_metadata.py    | Drop bogus 'notfound' metadata_overrides so AcoustID retries. --all   --since TS · --dry-run -y --log.                                                                                                                                                                                                                |
| T/a6  | relink_orphan_overrides.py    | Relink metadata_overrides orphaned by a UPnP rescan via d-id + fuzzy (artist,title). --apply --db.                                                                                                                                                                                                                    |
| T/a7  | relink_playlists_to_locals.py | Repoint playlist_tracks/favourites at RoHaLocalFS by normalised metadata; prune no-match. --apply --no-backup --no-prune-favs -y.                                                                                                                                                                                     |
| T/a8  | audit_override_mismatches.py  | Delete acoustid overrides whose artist AND title both mismatch the track (d-id collision repair). --clean --top -y.                                                                                                                                                                                                   |
| T/a9  | correct_year_drift.py         | Rewrite metadata_overrides.year to the earliest plausible year evidenced by another copy in the library. --apply --top --db -y.                                                                                                                                                                                       |
| T/a10 | improve_song_years.py         | Query MusicBrainz for each song's earliest recording year → song_year_cache → apply. --lookup --apply --limit -v.                                                                                                                                                                                                     |
| T/a11 | locals_scan.py                | Standalone LocalFs indexer (CLI). NOTE: do not run against the live DB without a base_url — leaves placeholder URLs.                                                                                                                                                                                                  |
| T/a12 | locals_serve.py               | Standalone LocalFs file-server launcher for testing the :8200 serving path.                                                                                                                                                                                                                                           |
| T/a13 | beets_enrich.py               | Run the beets tag-in-place enrichment batch (docs/enrichment.md). Safe wrapper around `beet import`: enforces write:yes/copy:no/move:no before any write. --write-config --quiet --timid --album --revisit --reindex --gateway --dry-run -y.                                                                          |
| T/a14 | post_beets_reindex.py         | The post-beets step: drop AcoustID metadata_overrides (LocalFs URLs are path-stable, so old acoustid rows would re-mask beets' fresh tags) then reindex LocalFs. Refuses to clear while ACOUSTID_API_KEY is set. --apply --dry-run --no-clean --no-reindex --no-backup --ignore-acoustid-key --udn --gateway --db -y. |

LAN devices (D/\*), external services (E/\*), scheduled jobs (J/\*)

| Code | Name                                                    | Role / protocol                                                                                                                                                                                                                   |
|------|---------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D/1  | Naim Uniti                                              | UPnP MediaRenderer. Receives AVTransport+RenderingControl SOAP (control/volume); pulls audio bytes over HTTP Range from :8200; also browses gateway playlists as a control point. Gateway is NOT in its audio path → bit-perfect. |
| D/2  | UPnP MediaServer                                        | MinimServer / generic UPnP (AssetUPnP decommissioned). Browsed via P/g10 UpnpProvider + P/g13 ContentDirectory SOAP. Kept for any non-LocalFs source.                                                                             |
| D/3  | Music files                                             | /Volumes/SAMDATA/Music (external drive, read-only). Indexed by P/g9; tags read by locals provider; bytes served by P/g11.                                                                                                         |
| D/4  | fpcalc                                                  | Chromaprint CLI binary (brew). Local fingerprinter for P/g18. Internal/local — not network.                                                                                                                                       |
| E/x1 | musicbrainz.org                                         | GET /ws/2/release-group — MBID + original year. UA + 1.1 s rate limit required.                                                                                                                                                   |
| E/x2 | coverartarchive.org                                     | HEAD /release-group/{mbid}/front-500 — cover presence.                                                                                                                                                                            |
| E/x3 | lrclib.net                                              | GET /api/get — on-demand lyrics for the playing track.                                                                                                                                                                            |
| E/x4 | *.api.radio-browser.info                                | GET /json/stations/search — internet-radio catalogue (HLS filtered).                                                                                                                                                              |
| E/x5 | api.acoustid.org                                        | POST /v2/lookup — fingerprint → MusicBrainz metadata. Needs an APPLICATION key.                                                                                                                                                   |
| J/1  | com.roha.dlna-gateway                                   | LaunchAgent that runs the gateway. Restart: launchctl kickstart -k gui/\${id -u}/com.roha.dlna-gateway.                                                                                                                           |
| J/2  | com.roha.dlna-cert-renew + renew-cert.sh                | Weekly TLS cert renewal (Mon 04:30; no-op unless <30 days). cert-renewal.log.                                                                                                                                                     |
| J/3  | com.roha.dlna-acoustid-retry + retry-acoustid-weekly.sh | Weekly notfound-clear + gateway restart for AcoustID (Mon 05:30). DISABLED under Option A (beets is the metadata authority) — agent unloaded via launchctl bootout. acoustid-retry.log.                                           |
| J/4  | setup.sh                                                | venv setup + run / restart / probe. --run --restart --no-browser --debug --probe URL --list-devices --reset-devices. See the setup.sh options reference.                                                                          |

Commands & options used in the drawing

| Context              | Command                                                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------------------------------------|
| Run (J/4)            | ./setup.sh --run [--no-browser] [--debug] [--probe http://...] [--list-devices] [--reset-devices]                        |
| Restart (J/4)        | ./setup.sh --restart (refresh venv/deps + launchctl kickstart -k gui/\${id -u})/com.roha.dlna-gateway)                   |
| Cert (J/2)           | ./renew-cert.sh [--force] · launchctl kickstart gui/\${id -u})/com.roha.dlna-cert-renew                                  |
| AcoustID retry (J/3) | ./retry-acoustid-weekly.sh [--dry-run]                                                                                   |
| Subsonic auth        | launchctl setenv SUBSONIC_PASSWORD ... ; launchctl getenv SUBSONIC_PASSWORD                                              |
| Full test suite      | python tests/run_all.py [--offline   --frontend   --frontend-only   http://host:8765]                                    |
| Unit / Playwright    | python3 -m unittest tests.test_player ... ; .venv/bin/pytest tests/frontend -v                                           |
| Chaos (live)         | python3 tests/chaos.py --iterations N --workers M [--seed S] [--quiet] [--base https://host:8443]                        |
| Schema gate (T/a1)   | python3 tools/regen_schema.py [--check]                                                                                  |
| Module self-tests    | python dlna_discovery.py   dlna_content.py   dlna_library.py   db_pool.py   dlna_player.py   dlna_server.py              |
| Range / 206 check    | curl -r 0-1023 -D - http://<host>:8200/localhosts/stream/<id> -o /dev/null                                               |
| Tools (T/a2–a12)     | python3 tools/<tool>.py [--dry-run --apply --trash --hard-delete --limit N --exts a,b -v -y] (see the tools table above) |

Ports (2.0): 8443 HTTPS — Hypercorn TLS + HTTP/2 (ALPN), tailscale cert · 8765 plain HTTP · 8200 RoHaLocalFS file server · 8770 device-tier /gw/\* (plain, for the Naim; SSDP advert points here) · 26125 (legacy AssetUPnP, decommissioned). The HTTP/2 + app-owned-TLS roadmap is now DONE — Hypercorn terminates TLS/h2 natively. Cutover: docs/CUTOVER\_RUNBOOK.md + CUTOVER\_LAUNCHD.md.

Tool options reference — what each flag does

Every option for every maintenance tool, with its meaning. Codes match the diagram (T/aN) and the tools table. Unless noted, file-deleting tools default to dry-run / report-only and move to the macOS Trash (recoverable ~30 days) — **--hard-delete** is the permanent, non-recoverable variant.

T/a1 regen\_schema.py — Regenerate the committed schema.sql.

| Option    | Meaning                                                                        |
|-----------|--------------------------------------------------------------------------------|
| (no args) | Rewrite schema.sql from the live LibraryDB schema.                             |
| --check   | Don't write; exit non-zero if schema.sql is stale (the test_schema_sync gate). |

T/a2 prune\_empty\_music\_dirs.py — Trash directories whose subtree has zero music files.

| Option         | Meaning                                                                                                  |
|----------------|----------------------------------------------------------------------------------------------------------|
|                | Music root to walk (positional).                                                                         |
| --dry-run      | Print decisions; touch nothing.                                                                          |
| -v / --verbose | Also log every KEPT directory (default: only deletions print).                                           |
| --limit N      | Stop after evaluating N dirs; when the limit is hit, NO deletions run (a partial picture = safety belt). |
| --exts a,b,c   | Override the music-extension set (commas, dot optional).                                                 |
| --hard-delete  | Permanent rm -rf instead of Trash. NOT recoverable.                                                      |
| -y / --yes     | Skip the confirmation prompt.                                                                            |

T/a3 find\_corrupt\_audio.py — Flag files whose magic bytes are wrong/zero for their extension.

| Option         | Meaning                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------|
|                | Music root to scan (positional).                                                                 |
| -v / --verbose | Also log every OK file (default: only corrupt).                                                  |
| --limit N      | Stop after scanning N files (0 = no limit); halts the delete step when hit.                      |
| --exts a,b,c   | Override the audio-extension list.                                                               |
| --out PATH     | Where to write the corrupt-paths list (default ./corrupt-audio.txt; pass /dev/null to suppress). |
| --trash        | Move flagged files to the Trash.                                                                 |
| --hard-delete  | Permanent unlink instead of Trash. Mutually exclusive with --trash.                              |
| -y / --yes     | Skip the confirmation prompt before deleting.                                                    |

T/a4 find\_duplicate\_audio.py — Find duplicate recordings on disk (same acoustid metadata); keeps a ranked winner.

| Option         | Meaning                                          |
|----------------|--------------------------------------------------|
|                | Music root to scan (positional).                 |
| -v / --verbose | Per-URL ambiguity / not-found logs.              |
| --trash        | Move the loser files to the Trash (winner kept). |
| --hard-delete  | Permanent delete of losers. NOT recoverable.     |
| -y / --yes     | Skip the confirmation prompt.                    |

T/a5 retry\_notfound\_metadata.py — Drop bogus 'notfound' metadata\_overrides so AcoustID retries them.

| Option       | Meaning                                                                               |
|--------------|---------------------------------------------------------------------------------------|
| (no args)    | Report only: counts + HTTP 5xx lines found in the log. No deletions.                  |
| --all        | Delete EVERY source='notfound' row (re-runs legit misses too).                        |
| --since 'TS' | Delete only notfound rows updated after the timestamp. Mutually exclusive with --all. |
| --dry-run    | Show what would be deleted without acting.                                            |
| --log PATH   | Log file to scan for HTTP 5xx (default acoustid-firstpass.log).                       |
| -y / --yes   | Skip the confirmation prompt.                                                         |

T/a6 relink\_orphan\_overrides.py — Relink metadata\_overrides orphaned by a UPnP rescan (d-id + fuzzy artist/title).

| Option    | Meaning                        |
|-----------|--------------------------------|
| (default) | Dry-run preview — no mutation. |
| --apply   | Perform the relinks.           |
| --db PATH | library.db path.               |

T/a7 relink\_playlists\_to\_locals.py — Repoint playlists/favourites at RoHaLocalFS by normalised metadata.

| Option          | Meaning                                                       |
|-----------------|---------------------------------------------------------------|
| (default)       | Dry-run preview.                                              |
| --apply         | Commit the relinks (auto-backs up library.db first).          |
| --no-backup     | Skip the automatic library.db backup on --apply.              |
| --no-prune-favs | Keep album_favourites that no longer match any LocalFs album. |
| -y / --yes      | Skip the confirmation prompt.                                 |

T/a8 audit\_override\_mismatches.py — Delete acoustid overrides whose artist AND title both mismatch the track.

| Option     | Meaning                                    |
|------------|--------------------------------------------|
| (default)  | Dry-run, lists the top 30 suspects.        |
| --clean    | Delete the suspect acoustid override rows. |
| --top N    | Show N rows (0 = full list).               |
| -y / --yes | Non-interactive (skip the confirm).        |

T/a9 correct\_year\_drift.py — Rewrite metadata\_overrides.year to the earliest plausible year from another copy in the library.

| Option     | Meaning                                     |
|------------|---------------------------------------------|
| (default)  | Dry-run, top 30 candidates.                 |
| --apply    | Write the corrections (as source='manual'). |
| --top N    | Preview N candidates (0 = all).             |
| --db PATH  | library.db path.                            |
| -y / --yes | Non-interactive apply.                      |

T/a10 improve\_song\_years.py — Query MusicBrainz for each song's earliest recording year → cache → apply.

| Option         | Meaning                                                                     |
|----------------|-----------------------------------------------------------------------------|
| (default)      | Dry-run preview — no MB calls, no DB writes.                                |
| --lookup       | Query MB for uncached (artist,title) groups; fill song_year_cache (sticky). |
| --apply        | Write cached hits onto tracks (metadata_overrides.year, source='manual').   |
| --limit N      | Cap the number of groups queried (for testing).                             |
| -v / --verbose | Per-group logging.                                                          |

T/a11 localsfs\_scan.py — Standalone LocalFs indexer (CLI). Don't run against the live DB without a base\_url — it leaves placeholder URLs.

| Option         | Meaning                                                                         |
|----------------|---------------------------------------------------------------------------------|
| --root PATH    | Music root to scan (default \$LOCALFS_MUSIC_ROOT or /Volumes/SAMDATA/Music).    |
| --force        | Ignore the (mtime, size) cache and re-tag every file. Use after schema changes. |
| --compare      | Skip scanning — just print tracks/albums per UDN currently in library.db.       |
| --db PATH      | library.db path.                                                                |
| -v / --verbose | Per-file logging.                                                               |

T/a12 localsfs\_serve.py — Standalone RoHaLocalFS file-server launcher (test the :8200 path).

| Option         | Meaning                                                           |
|----------------|-------------------------------------------------------------------|
| --port N       | HTTP port (default 8200 / \$LOCALFS_PORT).                        |
| --host ADDR    | Listen address (default 0.0.0.0 so the Naim can reach it).        |
| --db PATH      | library.db path.                                                  |
| --root PATH    | Allowed music root; repeatable. Defaults to \$LOCALFS_MUSIC_ROOT. |
| -v / --verbose | Request logging.                                                  |



T/a13 `beets_enrich.py` — Run the beets tag-in-place enrichment batch (docs/enrichment.md). Safe wrapper around ``beet import``. Deps: brew install chromaprint; pip3 install beets pyacoustid musicbrainzngs (beets 2.x pluginized MusicBrainz — the musicbrainz plugin + musicbrainzngs are REQUIRED or it matches nothing).

| Option               | Meaning                                                                                                                                                  |
|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| (no --quiet/--timid) | Interactive: beets prompts you per album (apply / skip / ...); strong matches auto-apply.                                                                |
| --write-config       | Write the prog-tuned tag-in-place ~/.config/beets/config.yaml (enables the musicbrainz metadata plugin; backs up any existing) and exit. Run this first. |
| --music-root PATH    | Library root to import (default /Volumes/SAMDATA/Music).                                                                                                 |
| --album PATH         | Import a single album directory instead of the whole root.                                                                                               |
| --config PATH        | beets config path (default ~/.config/beets/config.yaml).                                                                                                 |
| --quiet              | Unattended BULK pass: auto-accept only strong matches (≥ strong_rec_thresh 0.80), skip the rest, no prompts (beet import -q).                            |
| --timid              | Most cautious: prompt for EVERY match (per-change review). Mutually exclusive with --quiet.                                                              |
| --revisit            | Re-import a directory beets already recorded as done (-I / noincremental). Use after re-tagging an album.                                                |
| --reindex            | After import, find the LocalFs server UDN via /api/servers and POST /api/index/rebuild so the gateway picks up the new tags.                             |
| --gateway URL        | Gateway base URL for --reindex (default http://127.0.0.1:8765).                                                                                          |
| --udn UDN            | Explicit server UDN to reindex (default: auto-pick the uuid:localfs-* server).                                                                           |
| -n / --dry-run       | Print the resolved beet command + safety report; do NOT invoke beets (beets has no true dry-run of its own).                                             |
| -y / --yes           | Skip the in-place-write backup-warning confirmation.                                                                                                     |

T/a14 `post_beets_reindex.py` — After beets has tagged files in place, make its work visible: clear the AcoustID metadata\_overrides (LocalFs URLs are PATH-stable, so the COALESCE pass would otherwise re-lay old acoustid rows over beets' fresh tags) THEN reindex LocalFs. source='manual' is never touched; notfound/video\_skip carry NULL metadata so they mask nothing.

| Option                | Meaning                                                                                                                                                                                                                                                |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (default)             | Dry-run: print the override breakdown + the planned clear/reindex; change nothing.                                                                                                                                                                     |
| --apply               | Actually delete acoustid overrides + start the reindex (auto-backs up library.db first).                                                                                                                                                               |
| -n / --dry-run        | Explicit preview alias (the default when --apply is absent).                                                                                                                                                                                           |
| --no-clean            | Skip clearing overrides (reindex only).                                                                                                                                                                                                                |
| --no-reindex          | Skip the reindex (clean only).                                                                                                                                                                                                                         |
| --no-backup           | Skip the library.db backup before deleting.                                                                                                                                                                                                            |
| --ignore-acoustid-key | Proceed even if ACOUSTID_API_KEY is set. By default the tool REFUSES to clear while the worker is live — the 120s startup scan would re-fingerprint every now-bare track and re-create the overrides, re-masking beets (Option A keeps the key unset). |
| --udn UDN             | Server UDN to reindex (default: auto-pick uuid:localfs-*).                                                                                                                                                                                             |
| --gateway URL         | Gateway base URL (default http://127.0.0.1:8765).                                                                                                                                                                                                      |
| --db PATH             | library.db path.                                                                                                                                                                                                                                       |
| -y / --yes            | Skip the confirmation prompt.                                                                                                                                                                                                                          |

J/4 `setup.sh` — Bootstrap + run the gateway. Finds Python 3.9+, creates/repairs the .venv, installs requirements.txt, then either runs the gateway (--run, exec in the foreground), restarts the launchd-managed copy (--restart), or just finishes setup. Unknown flags after --run are forwarded verbatim to dlina\_gateway.py.

| Option             | Meaning                                                                                                                                                                                                                                                                                                                    |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (no flags)         | Set up only: venv + dependencies, then print the 'how to run' summary. Does not start anything.                                                                                                                                                                                                                            |
| --run              | Set up, then exec the gateway in the foreground on :8765 (blocks; Ctrl-C to stop). All flags below it are passed through to dlina_gateway.py.                                                                                                                                                                              |
| --restart          | Refresh the venv/deps, then restart the launchd gateway via launchctl kickstart -k gui/\$(id -u)/com.roha.dlina-gateway (the launchd-correct restart — a bare kill races launchd's respawn). Aborts with install hints if the LaunchAgent isn't loaded. Mutually informative with --run; --restart wins if both are given. |
| --no-browser       | (forwarded) Don't auto-open the browser on start.                                                                                                                                                                                                                                                                          |
| --debug            | (forwarded) Verbose logging.                                                                                                                                                                                                                                                                                               |
| --probe http://... | (forwarded) Add a UPnP server manually by its device-description URL.                                                                                                                                                                                                                                                      |
| --list-devices     | (forwarded) Print the known-devices table and exit.                                                                                                                                                                                                                                                                        |
| --reset-devices    | (forwarded) Wipe the device DB and exit.                                                                                                                                                                                                                                                                                   |
| --port N           | (forwarded) HTTP port (default 8765).                                                                                                                                                                                                                                                                                      |